Francesco **Salvi**

Research Fellow · EPFL

he/him | salvi.fnc@gmail.com | frasalvi.github.io | frasalvi | frasalvi | fraslv | Scholar

August 31, 2025

Princeton University
Application as PhD student in Computer Science

Statement of Purpose

My research interests lie at the intersection of **Computational Social Science**, **Natural Language Processing**, and **Human-Computer Interaction**. I am particularly invested in studying the **societal impacts of LLMs and agentic systems**, leveraging tools from causal inference and network science to draw insights into human behavior. During my studies at EPFL, I had the opportunity to engage in research across five different academic laboratories and two internships in industry, leading to 4 research papers published or currently under review at leading conferences and journals. My desire to pursue a PhD stems from a deep commitment to ensuring that artificial intelligence serves humanity responsibly, with a long-term goal of building a career in academia.

Research experience. My interest in research comes from a variety of positive experiences over the last four years. After a successful project conducted during my first semester at EPFL, placing me in the top 5% in a 440+ students course, I was selected to join Prof. Robert West's *Data Science Lab*. There, I participated in an ambitious collaboration with the Wikimedia Foundation to study how images are used on Wikipedia, characterizing user interaction with visual content through a novel classifier. Inspired by this experience, I then joined the *Information and Network Dynamics Lab*. Under the supervision of Prof. Matthias Grossglauser, I studied the influence of interest groups on members of the European Parliament. Using NLP techniques, we uncovered hidden links between parliamentarians and lobbies by comparing their speeches and position papers, resulting in a publication at **NAACL 2024 (Main)** [1]. Delving deeper into the fundamentals of NLP, I also spent a summer between the *NLP Lab* and the *ML & Optimization Lab*, where I worked with Prof. Antoine Bosselut and Prof. Martin Jaggi in a vast collaborative effort to train large-scale language and multimodal models adapted for clinical practice. Our MEDITRON-70B [2] achieved **state-of-the-art performance** among open models; its artifact has been **downloaded over 100k times** and our paper, under review at **Nature Medicine**, has been **cited over 200 times** in less than a year.

The heterogeneity of these experiences taught me a lot about research and exposed me to the latest advances in a variety of subfields. By experiencing different ways of doing science, I also realized that my true vocation lies in **research that connects computational tools to society**, aiming to understand individual and collective behavior and the societal implications of new technologies. For my Master's Thesis, therefore, I visited the *Complex Human Behavior Lab* at the Bruno Kessler Foundation (FBK), led by Dr. Riccardo Gallotti, where I designed a randomized controlled trial to measure the persuasiveness of LLMs. Our research [3], currently under review at **Nature Human Behavior** (Altmetric Attention Score >1000, 99.99th percentile compared to outputs of similar age), found that LLMs outperform humans in real-time online debates, especially when provided with

the sociodemographics of their debating partner. Our work received extensive coverage by various media outlets, including **The Economist**, The Register, New Scientist, and programs on Swiss National TVs and radios. Additionally, it **informed Anthropic's model evaluation strategy** [4].

Future Research Agenda. For my PhD, I intend to continue exploring the emerging capabilities of LLMs and agentic systems. Recent advances in such technologies simultaneously present serious threats to online communities and offer promising opportunities to enhance their well-being. To disentangle these two aspects, I will focus on a prominent issue that exemplifies the intertwined risks and benefits of AI: political polarization. Online polarization has long been recognized for exacerbating societal divisions and fragmentation, posing a significant threat to the stability of our democracies [5, 6]. LLMs could potentially escalate such risks to unprecedented levels, intensifying the rise of polarization by persuading individuals to adopt radical beliefs through personalized messages. Conversely, LLM-powered agents could enable more effective interventions in contexts where traditional methods have largely failed [7, 8], enhancing content moderation and improving online conversations.

To understand RQ1 **how LLMs contribute to online polarization and political dysfunction**, it is crucial to first assess the capabilities of frontier models and understand how they are being exploited by malicious actors online. Moving beyond the classical framework of artificial lab experiments, my approach will combine observational studies and field experiments to measure the persuasiveness of LLMs in spontaneous conversations online. I am already taking the first steps in this direction by developing LLM bots for Reddit communities and by co-leading a large team of 50+ interdisciplinary researchers focused on persuasion research. In parallel, I plan to use instrumental variables such as API outages and other proxies for LLM usage to study their proliferation on social media, measuring the extent to which interactions with AI shape political behaviors and foster polarization across different groups.

Building on these results, I will investigate RQ2 **how LLMs can be leveraged to reduce polarization in online communities** by developing targeted interventions to mitigate political dysfunction. Drawing inspiration from the growing body of literature on simulating human behavior with LLMs [9–11] and creating agent-based simulations of social networks [12, 13], I will employ synthetic simulacra to iterate over the large space of potential interventions. These may include re-ranking social feeds to improve social media algorithms [14] and developing personalized assistants to encourage respectful and empathetic interactions. The most promising interventions will then be tested at scale on online platforms to generate actionable recommendations that can inform policy regulations and improve the next generation of social media.

Why Princeton? At Princeton, I would be particularly excited to work with **Prof. Manoel Horta Ribeiro** because of his extensive expertise with observational studies and his recent work on the impact of generative AI. I am also genuinely captivated by **Prof. Aleksandra Korolova**'s eclectic research on algorithmic fairness and auditing and **Prof. Arvind Narayanan**'s work on AI ethics. More broadly, I would be thrilled to join Princeton's vibrant and interdisciplinary community, where I could join forces with talented students to help shape the future of social computing. After my PhD, I hope to pursue a career in academia as a professor, raising a strong voice to ensure that advancements in AI and technology benefit society in an aligned, inclusive, and responsible manner.

References

[1] Aswin Suresh, Lazar Radojević*, **Francesco Salvi***, Antoine Magron, Victor Kristof, and Matthias Grossglauser. "Discovering Lobby-Parliamentarian Alignments through NLP". In: *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Ed. by Kevin Duh, Helena Gomez, and Steven Bethard. Mexico City, Mexico: Association for Computational Linguistics, 2024, pp. 8107–8120.

[2] Zeming Chen, Alejandro Hernández Cano, Angelika Romanou, Antoine Bonnet, Kyle Matoba, **Francesco Salvi**, Matteo Pagliardini, Simin Fan, Andreas Köpf, Amirkeivan Mohtashami, Alexandre Sallinen, Alireza Sakhaeirad, Vinitra Swamy, Igor Krawczuk, Deniz Bayazit, Axel Marmet, Syrielle Montariol, Mary-Anne Hartley, Martin Jaggi, and Antoine Bosselut. *MEDITRON-70B: Scaling Medical Pretraining for Large Language Models*. 2023.

[3] **Francesco Salvi**, Manoel Horta Ribeiro, Riccardo Gallotti, and Robert West. *On the Conversational Persuasiveness of Large Language Models: A Randomized Controlled Trial*. 2024.

[4] Esin Durmus, Liane Lovitt, Alex Tamkin, Stuart Ritchie, Jack Clark, and Deep Ganguli. *Measuring the Persuasiveness of Language Models*. Apr. 9, 2024. URL: `https://www.anthropic.com/news/measuring-model-persuasiveness`.

[5] Joshua Tucker, Andrew Guess, Pablo Barbera, Cristian Vaccari, Alexandra Siegel, Sergey Sanovich, Denis Stukal, and Brendan Nyhan. "Social Media, Political Polarization, and Political Disinformation: A Review of the Scientific Literature". In: *SSRN Electronic Journal* (2018). ISSN: 1556-5068.

[6] Jonathan Haidt and Christopher Bail. *Social Media and Political Dysfunction: A Collaborative Review*. 2023.

[7] Christopher A. Bail, Lisa P. Argyle, Taylor W. Brown, John P. Bumpus, Haohan Chen, M. B. Fallin Hunzaker, Jaemin Lee, Marcus Mann, Friedolin Merhout, and Alexander Volfovsky. "Exposure to opposing views on social media can increase political polarization". In: *Proceedings of the National Academy of Sciences* 115.37 (2018), pp. 9216–9221.

[8] Andrew M. Guess, Neil Malhotra, Jennifer Pan, Pablo Barberá, Hunt Allcott, Taylor Brown, Adriana Crespo-Tenorio, Drew Dimmery, Deen Freelon, Matthew Gentzkow, Sandra González-Bailón, Edward Kennedy, Young Mie Kim, David Lazer, Devra Moehler, Brendan Nyhan, Carlos Velasco Rivera, Jaime Settle, Daniel Robert Thomas, Emily Thorson, Rebekah Tromble, Arjun Wilkins, Magdalena Wojcieszak, Beixian Xiong, Chad Kiewiet de Jonge, Annie Franco, Winter Mason, Natalie Jomini Stroud, and Joshua A. Tucker. "How do social media feed algorithms affect attitudes and behavior in an election campaign?" In: *Science* 381.6656 (July 2023), pp. 398–404. ISSN: 1095-9203.

[9] Joon Sung Park, Carolyn Q. Zou, Aaron Shaw, Benjamin Mako Hill, Carrie Cai, Meredith Ringel Morris, Robb Willer, Percy Liang, and Michael S. Bernstein. *Generative Agent Simulations of 1, 000 People*. 2024.

[10] Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. "Out of One, Many: Using Language Models to Simulate Human Samples". In: *Political Analysis* 31.3 (Feb. 2023), pp. 337–351. ISSN: 1476-4989.

[11] Luke Hewitt, Ashwini Ashokkumar, Isaias Ghezae, and Robb Willer. *Predicting Results of Social Science Experiments Using Large Language Models*. 2024.

[12] Petter Törnberg, Diliara Valeeva, Justus Uitermark, and Christopher Bail. *Simulating Social Media Using Large Language Models to Evaluate Alternative News Feed Algorithms*. 2023.

[13] Joon Sung Park, Joseph O'Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. "Generative Agents: Interactive Simulacra of Human Behavior". In: *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. UIST '23. San Francisco, CA, USA: Association for Computing Machinery, 2023. ISBN: 9798400701320.

[14] Tiziano Piccardi, Martin Saveski, Chenyan Jia, Jeffrey T. Hancock, Jeanne L. Tsai, and Michael Bernstein. *Social Media Algorithms Can Shape Affective Polarization via Exposure to Antidemocratic Attitudes and Partisan Animosity*. 2024.